Vector Fields and Line Integrals

Section 1: Vector Fields, Divergence, and Curl

1 Compute the divergence $\nabla \cdot \mathbf{F}$ for $\mathbf{F} = \langle x^2 y,\ 3yz,\ -2xz \rangle$.

a. $2xy + 3z - 2x$

b. $2xy + 3y - 2z$

c. $x^2 + 3z - 2x$

d. 0

2 Compute the 2D curl of $\mathbf{F} = \langle -y, x \rangle$.

a. 2

b. 0

c. 1

d. -2

3 If a vector field has $\nabla \times \mathbf{F} = \mathbf{0}$ throughout a simply connected domain, what is it called?

a. Conservative (or irrotational)

b. Incompressible

c. Solenoidal

d. Harmonic

Section 2: Line Integrals with Respect to Arc Length

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4 State the formula for $\int_C f(x,y)\,ds$ along a curve parameterized by $\mathbf{r}(t)$ for $a \le t \le b$.

a. $\int_a^b f(x(t),y(t))\sqrt{[x'(t)]^2+[y'(t)]^2}\,dt$

b. $\int_a^b f(x(t),y(t))\,dt$

c. $\int_a^b f(x(t),y(t))[x'(t)+y'(t)]\,dt$

d. $\int_a^b f'(t)\,dt$

5 Evaluate $\int_C (x+y)\,ds$ where C is the line segment from $(0,0)$ to $(3,4)$.

a. $\frac{35}{2}$

b. 35

c. 7

d. 12

6 Find the mass of a wire bent in the shape of the semicircle $\mathbf{r}(t)=\langle 2\cos t, 2\sin t\rangle$ $(0 \le t \le \pi)$ with density $\rho(x,y)=y$.

a. 4

b. 8

c. 2π

d. 4π

Section 3: Line Integrals of Vector Fields (Work Integrals)

7 Evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$ for $\mathbf{F}=\langle y,\,-x\rangle$ along the circle $\mathbf{r}(t)=\langle \cos t, \sin t\rangle$ for $0 \le t \le 2\pi$.

a. -2π

b. 2π

c. 0

d. π

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8 Find the work done by the constant force $\mathbf{F} = \langle 3, 5 \rangle$ moving an object along the straight line from $(1, 1)$ to $(4, 5)$.

a. 29

b. 24

c. 32

d. 15

9 Evaluate $\int_C (x\,dx + y\,dy)$ along the parabolic arc $y = x^2$ from $(0, 0)$ to $(1, 1)$.

a. 1

b. 1/2

c. 2/3

d. 0

Section 4: Fundamental Theorem of Line Integrals and Green's Theorem

10 State the Fundamental Theorem of Line Integrals for a conservative field $\mathbf{F} = \nabla f$.

a. $\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$

b. $\int_C \nabla f \cdot d\mathbf{r} = 0$ for all curves

c. $\int_C \nabla f \cdot d\mathbf{r} = \|\nabla f\|$

d. $\int_C \nabla f \cdot d\mathbf{r} = f'(\mathbf{r}(b))$

11 State Green's Theorem relating a line integral along a positively oriented boundary ∂D to a double integral over D.

a. $\oint_C (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$

b. $\oint_C (P\,dx + Q\,dy) = \iint_D (P + Q)\,dA$

c. $\oint_C (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right) dA$

d. $\oint_C (P\,dx + Q\,dy) = 0$

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12 Use Green's Theorem to evaluate $\oint_C (3y\,dx + 5x\,dy)$ where C is the unit circle $x^2 + y^2 = 1$ oriented counterclockwise.

a. 2π

b. 8π

c. 0

d. π

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